Week 1

Project Setup & Dataset Preparation

- Finalize system architecture
- Set up project structure
- Configure GitHub repository
- Create Python virtual environment
- Set up FastAPI and frontend (Streamlit/React)
- Download legal datasets (Acts, judgments, contracts)
- Organize raw and processed data folders

Week 2

Document Ingestion & Parsing Pipeline

Goal

- Extract clean text from multiple legal document formats.

Tasks

- Build PDF parser using PyMuPDF
- Build HTML parser
- Add OCR support for scanned PDFs
- Extract metadata (court, date, judge, section)
- Detect document type
- Remove duplicate documents
- Store cleaned documents

Week 3

Legal Text Cleaning & Hierarchy-Aware Chunking

Goal

- Convert raw legal text into intelligent chunks.

Tasks

- Remove headers, footers, and page numbers
- Normalize text
- Preserve legal formatting
- Build hierarchy-aware chunker
- Preserve chapters, sections, subsections
- Attach metadata to every chunk
- Store processed chunks

Week 4

Embedding Pipeline & Vector Database

Goal

- Create searchable semantic representations.

Tasks

- Generate embeddings
- Implement embedding cache
- Batch embedding pipeline
- Set up Qdrant/PGVector
- Store vectors with metadata
- Create indexing pipeline

Week 5

Retrieval Pipeline & LLM Integration

This is where the complete RAG pipeline is built:

- User enters a legal query.
- Convert the query into an embedding.
- Retrieve relevant chunks from the vector database.
- Perform BM25 retrieval.
- Merge dense and sparse results.
- Send the top results to the reranker.

Build the final prompt with:

- Retrieved legal context
- User question
- IRAC instructions
- Send the prompt to the LLM (Claude/GPT/Gemini).
- Receive the generated answer.
- Display the answer.

Week 6

Cross-Encoder Reranking & Query Routing

Goal

- Retrieve the most relevant legal context.

Tasks

- Implement reranking model
- Rank retrieved documents
- Build query classifier
- Route queries to different workflows

- Create task-specific prompts

Week 7

Agentic RAG Workflow

Goal

- Transform the system into an Agentic AI application.

Tasks

- Build Router Agent
- Build Retrieval Agent
- Build Verification Agent
- Build Legal Reasoning Agent
- Build Formatting Agent
- Connect agents into one workflow

Week 8

Citation Verification & Hallucination Prevention

Goal

- Increase reliability and trustworthiness.

Tasks

- Build citation verifier
- Verify statutes and case citations
- Detect hallucinated citations
- Add confidence scoring
- Implement abstention mechanism

Week 9

Multi-Document Reasoning & Citation Graph

Goal

- Improve legal reasoning across multiple sources.

Tasks

- Build citation graph
- Link related judgments
- Implement precedent chaining
- Retrieve multiple related documents
- Build evidence aggregation module

Week 10

User Interface & Explainability

Goal

- Create a professional user experience.

Tasks

- Build chat interface
- Implement document upload
- Display retrieved sources
- Add explainability panel
- Show similarity scores
- Enable streaming responses
- Store chat history

Week 11

Evaluation Framework & Feedback Loop

Goal

- Measure and improve system quality.

Tasks

- Create golden QA dataset
- Integrate RAGAS
- Evaluate faithfulness
- Measure context precision and recall
- Build hallucination benchmark
- Add user feedback collection

Week 12

Production Features & Deployment

Goal

- Prepare the system for real-world use.

Tasks

- Implement JWT authentication
- Add rate limiting
- Secure file uploads
- Add logging and monitoring
- Dockerize the application
- Configure CI/CD
- Deploy to Railway/Render/AWS

Week 13

Testing, Optimization & Documentation

Goal

- Polish the project for interviews and your portfolio.

Tasks

- Perform end-to-end testing
- Optimize latency
- Improve retrieval performance
- Write comprehensive README
- Prepare architecture diagrams
- Record demo video
- Document evaluation results
- Prepare resume bullet points and presentation